

RAMAN INTERNATIONAL

MONTHLY TEST - ARTIFICIAL INTELLIGENCE

Name: _____

Board: CBSE

Roll No.: _____

Date: 1 September 2026

Subject: Artificial Intelligence

Time Allowed: 1.5 Hours

Exam: Monthly Test

Total Marks: 40

Pass Marks: 13

General Instructions:

1. All questions are compulsory.
2. Marks allotted to each question are indicated in square brackets [].
3. Read each question carefully before attempting it.
4. Write your answers clearly and legibly.
5. Do any rough work separately, not in the answer space.

SECTION A: VERY SHORT ANSWER QUESTIONS

[8 MARKS]

Question 1: Answer all 8 questions. Each carries 1 mark.

[8 × 1 = 8]

- a. What is the full form of **NLP** in the context of Artificial Intelligence?
- b. Name the first stage of the **AI Project Cycle**.
- c. Define '**Ethical Framework**' in the context of AI.
- d. *Fill in the blank:* Bioethics is a _____-based ethical framework primarily used in the healthcare sector.
- e. State the core ethical principle of '**Non-maleficence**' as applied in bioethics.
- f. Name the three primary domains into which AI models are broadly categorized based on the type of data they process.
- g. What does the **Deployment stage** of the AI Project Cycle ensure in a real-world environment?
- h. If an AI algorithm discriminates against a specific group of people in its recommendations, which principle of bioethics is being violated?

SECTION B: SHORT ANSWER QUESTIONS

[12 MARKS]

Question 1: Answer all 4 questions. Each carries 3 marks.

[4 × 3 = 12]

a. AI Project Cycle Data Stages:

- (i) Distinguish between the *Data Acquisition* and *Data Exploration* stages of the AI Project Cycle. Why is it important to represent acquired data visually?
- (ii) At which stage would you identify that your AI model is performing poorly on unseen data, and what action would you take next?
- (iii) Why is the AI Project Cycle described as 'cyclical' rather than 'linear'?

b. AI Domains (Computer Vision vs. NLP):

- (i) A smart traffic management system uses cameras to detect vehicles that jump red lights and automatically issues fines. Identify the AI domain being used and justify your answer.

- (ii) A chatbot on a banking website understands and responds to customer queries in Hindi. Identify the AI domain and state two key challenges this system might face.
- (iii) Can a single AI solution employ more than one domain simultaneously? Give one real-world example to support your answer.

c. Types of Ethical Frameworks:

- (i) Differentiate between *sector-based* and *value-based* ethical frameworks with one example for each.
- (ii) A self-driving car's AI must decide whether to swerve and harm one pedestrian or stay on course and harm three passengers in an unavoidable accident scenario. Which value-based ethical framework would evaluate this by calculating the greatest good for the greatest number? Explain its application here.
- (iii) How does a *virtue-based framework* differ from a *rights-based framework* when applied to AI development?

d. Bioethics and its Principles:

- (i) An AI diagnostic tool used in a hospital provides its risk predictions to doctors but does not disclose its underlying logic or training data to patients. Which principle of bioethics is being violated? Explain.
- (ii) Distinguish between '*Maleficence*' and '*Non-maleficence*' with an example from AI system design.
- (iii) How does the principle of '*Justice*' in bioethics require AI developers to go beyond simply avoiding harm in their systems?

SECTION C: APPLICATION, ANALYSIS & CASE-BASED QUESTIONS

[20 MARKS]

Question 1: Answer all 10 questions. Each carries 2 marks.

[10 × 2 = 20]

a. Recruitment Bias Scenario: An AI recruitment tool trained on historical hiring data consistently ranks male candidates higher than equally qualified female candidates.

- (i) Identify the stage in the AI Project Cycle where this bias most likely originated.
- (ii) Which AI domain is this tool primarily based on, and what type of data would it have been trained on?

b. Healthcare Allocation: A healthcare company develops an AI model to predict the likelihood of patients defaulting on their medical bill payments and uses this as a proxy for deciding the level of treatment priority.

- (i) Which principle of bioethics does this approach most severely violate, and why?
- (ii) Suggest one corrective measure using the principle of 'Beneficence'.

c. Assertion & Reasoning (Model Evaluation):

Assertion (A): An AI model evaluated successfully on training data may still fail in real-world deployment.

Reasoning (R): The Evaluation stage in the AI Project Cycle tests the model on newly fetched, unseen data to identify gaps before deployment.

- A) Both A and R are true, and R is the correct explanation of A.
- B) Both A and R are true, but R is not the correct explanation of A.
- C) A is true, but R is false.
- D) A is false, but R is true.

d. Assertion & Reasoning (Ethical Surveillance):

Assertion (A): A utility-based ethical framework would fully justify an AI surveillance system that monitors all citizens to prevent crime, as it maximizes safety for the majority.

Reasoning (R): Utility-based frameworks evaluate actions solely on the basis of maximizing overall good without any consideration for individual rights.

- A) Both A and R are true, and R is the correct explanation of A.
- B) Both A and R are true, but R is not the correct explanation of A.
- C) A is true, but R is false.
- D) A is false, but R is false.

e. Smart Agriculture: A farmer uses a drone-based AI system that captures aerial images of his fields and identifies patches where crops are diseased. The system then recommends targeted pesticide spraying only in affected zones.

(i) Identify the AI domain this system belongs to and name the process by which the drone images are converted into actionable recommendations.

(ii) Name one additional application of this AI domain in a completely different sector.

f. Project Cycle Structure: Draw a neat, labelled diagram of the AI Project Cycle showing all six stages in the correct cyclical sequence. Briefly explain why the arrow from the Evaluation stage can point back to the Modelling stage.

g. Language Barrier in Moderation: An NLP-based AI content moderation tool is trained on social media posts in English to detect hate speech. When deployed in India to moderate posts in regional languages, its accuracy drops significantly.

(i) At which stage of the AI Project Cycle should this limitation have been identified and addressed?

(ii) Apply any ONE principle of bioethics to argue why deploying this flawed tool causes harm.

h. Domain Classification: Classify the following AI applications into their correct AI domain (*Statistical Data, Computer Vision, or NLP*) and provide a one-line justification for each:

(i) A price prediction model for second-hand cars based on mileage, age, and service history.

(ii) A system that automatically generates subtitles for a video by processing the spoken audio.

(iii) A quality control system on a manufacturing line that uses cameras to detect defective products.

i. Case Scenario (Predictive Policing): A municipal corporation deploys an AI-based system to allocate police patrol resources across different neighbourhoods. The algorithm is trained on historical crime data. It consistently directs more police to economically weaker areas because those areas have historically had more recorded crime—in part because they were already over-policed. This creates a self-reinforcing feedback loop.

Question: Using any TWO principles of bioethics, critically evaluate whether this AI deployment is ethically justified. Suggest one concrete corrective measure for each principle you apply.

j. Case Scenario (EdTech Platform): An EdTech company launches an AI tutoring platform that adapts its teaching style and content based on each student's learning pace and performance data. The system collects detailed data on students' academic struggles, attention span patterns, and emotional responses (via facial expression analysis using computer vision). Parents and students are not explicitly informed about the extent of data being collected or how it is used to profile the student.

Question:

(i) Identify the TWO AI domains being used in this platform.

(ii) Analyse this scenario using the principle of 'Respect for Autonomy' from bioethics.

(iii) From a rights-based value framework perspective, state ONE right of the students that is being violated and justify your answer.